

In Situ Formation of a Lithiophilic Li–Zn Alloy Using Dual-Salt Electrolyte Additives for Lithium Metal Batteries

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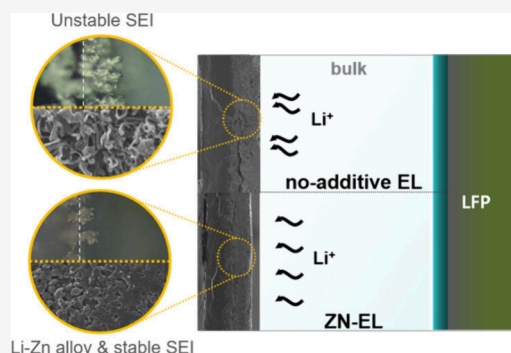
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ABSTRACT: The highly reactive nature of Li metal presents challenges, such as an unstable solid–electrolyte interphase (SEI) and uneven Li nucleation. Herein, to address these challenges, ZnF_2 and LiNO_3 were introduced as electrolyte additives to create a lithiophilic Li–Zn alloy and a stable SEI comprising LiF and LiNO_3 on the Li metal surface. The Li–Zn alloy, formed via a spontaneous conversion reaction between ZnF_2 and Li, provides favorable nucleation sites for Li deposition, thereby suppressing dendrite growth and improving the cycle stability of the Li metal, even at a high current density of 5 mA cm^{-2} . In Li||LiFePO_4 full cells, a remarkable capacity retention of 98.6% was achieved after 400 cycles at 1 C. Furthermore, approximately 100% capacity retention was achieved after 1400 cycles at 10 C. This study provides a simple and effective electrolyte additive strategy for obtaining dendrite-free Li metal.

KEYWORDS: Li metal batteries, dual-salt electrolyte, electrolyte additives, lithiophilic Li–Zn alloy, Li dendrite suppression



Although graphite anodes provide excellent conductivity when deployed in conventional Li-ion batteries,^{1,2} their low theoretical capacity (372 mAh g^{-1}) presents a drawback. Conversely, next-generation Li-metal batteries (LMBs) utilizing Li metals as anodes exhibit extremely high theoretical capacity (3860 mAh g^{-1}) and low electrochemical potential (-3.04 V vs SHE); they have emerged as representative candidate high-energy-density batteries.^{3,4} However, the high reactivity of Li metals accounts for several critical limitations in LMBs such as (1) interfacial side reactions with electrolytes, (2) uneven Li dendrite growth during repeated stripping/plating cycles, and (3) formation of dead Li separated from the dendrite matrix.^{5–8} In particular, the formation of Li dendrites must be suppressed to enable the practical application of LMBs, as dendrites directly impact battery performance.

Mechanically and electrochemically stable solid–electrolyte interphases (SEIs) can inhibit irreversible electrolyte loss resulting from parasitic reactions with Li metals, thereby suppressing dendrite growth. For example, a study revealed that LiF-rich SEIs with high mechanical strength suppressed crack formation caused by Li volume expansion.⁹ Additionally, electrochemically stable SEIs do not react with Li and electrolytes. Consequently, strategies, such as artificial SEI formation, adsorption of three-dimensional architectures onto Li metal, and electrolyte engineering, have been explored for designing highly stable SEIs.^{10–12} Additionally, electrolyte modification, which is among the simplest strategies for fabricating stable artificial SEIs, enables LiF-rich SEI formation

using high concentrations of lithium bis(fluorosulfonyl)imide (LiFSI)^{13,14} or additives, such as lithium nitrate (LiNO_3), lithium difluoro(oxalate)borate, fluoroethylene carbonate, and vinylene carbonate, that contribute to the N- and F-rich SEI layers after undergoing decomposition.^{15–20} Owing to their low interfacial energy, lithiophilic materials (e.g., Au, Ag, Zn, In, and Mg) reduce nucleation barriers and facilitate uniform Li deposition.^{21–23} As these materials exhibit the same structure as pure Li metal, they can improve the stability of Li growth, thereby effectively suppressing dendrite formation.^{21,23} Similarly, metal fluorides (e.g., AgF , AuF , InF_3 , MgF_2 , and ZnF_2) exhibit lithiophilicity toward Li and promote the formation of mechanically stable F[−]-rich SEIs. Among them, ZnF_2 is cost-effective, adsorbs onto the Li surface, and undergoes conversion reactions, thereby providing the fluoride required for SEI formation.

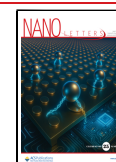
Herein, to simultaneously address the challenges of unstable SEI formation and nonuniform Li nucleation, we designed a Li metal surface with a lithiophilic layer and a mechanically and chemically stable SEI using ZnF_2 and LiNO_3 as electrolyte

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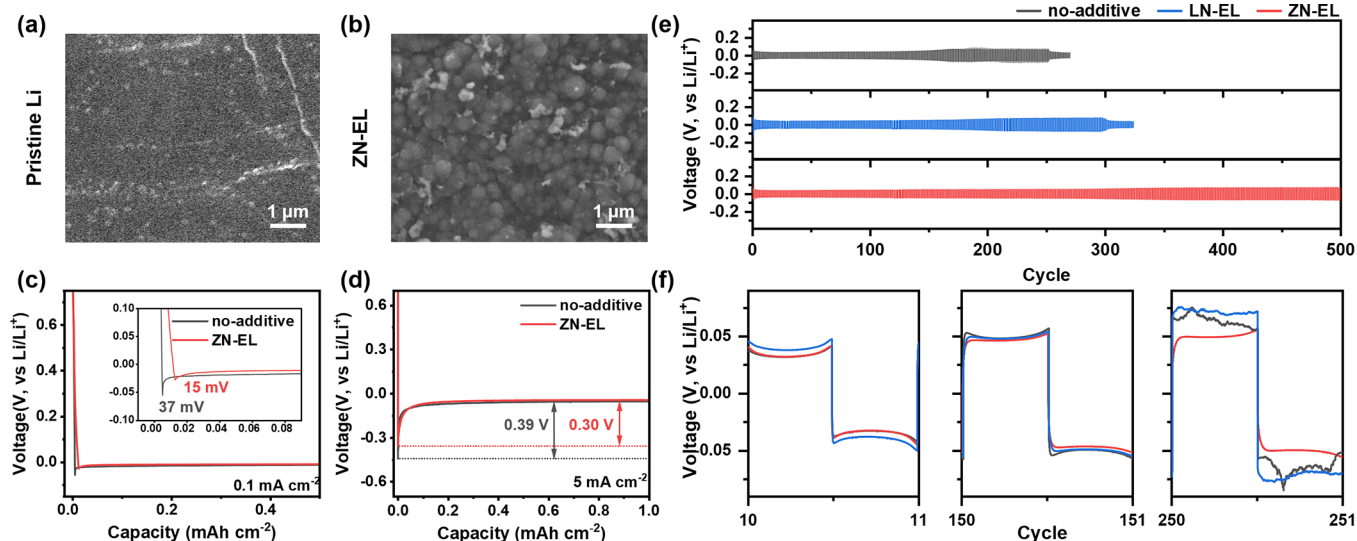


Figure 1. High-magnification FE-SEM images of the Li metal surface for (a) pristine Li and (b) after reaction with 2.3 M LiFSI in DME containing 40 mM ZnF_2 and 20 mM LiNO_3 (ZN-EL). Nucleation overpotential results on the Cu foil of a Li||Cu asymmetric cell at current densities of (c) 0.1 mA cm^{-2} and (d) 5 mA cm^{-2} . (e) Electrochemical performance of Li||Li symmetric cells at 5 mA cm^{-2} with a capacity set to 1 mAh cm^{-2} . (f) Corresponding enlarged views of the 11th, 150th, and 250th cycles.

additives. ZnF_2 undergoes a conversion reaction to form a Li–Zn alloy and LiF as follows:²²



During the reaction, the Li–Zn alloy forms on the Li metal surface, facilitating uniform Li plating/stripping. Although LiF demonstrates excellent mechanical strength, it exhibits low ionic conductivity, which can be compensated for by Li_3N formed from LiNO_3 .^{24–26} Thus, the formation of a lithiophilic Li–Zn layer promotes uniform Li nucleation, leading to granular growth and enhanced Li metal stability. Accordingly, herein, the additives extended the Li metal lifespan over 2-fold at 5 mA cm^{-2} in Li symmetric cells while also enhancing Coulombic efficiency (CE) and rate capability in Li||LiFePO₄ (LFP) full cells.

The initial uneven SEI formation caused Li dendrite growth, which negatively affected the performance and stability of the cell. ZnF_2 and LiNO_3 were introduced as electrolyte additives to form a reinforced SEI on the Li metal surface. The electrolytes were prepared using 2.3 M LiFSI in 1,2-dimethoxyethane (DME) (no-additive EL, where EL denotes ‘electrolyte’), containing 20 mM LiNO_3 (N-EL), 80 mM LiF and 20 mM LiNO_3 (LN-EL), and 40 mM ZnF_2 and 20 mM LiNO_3 (ZN-EL) depending on the compositions of the additives, and stirred for 3 d until additives were completely dispersed. After the Li metal was exposed to each electrolyte, the reacted surface was analyzed via field-emission scanning electron microscopy (FE-SEM). Compared with pristine Li metal (Figure 1a), spherical particles were uniformly distributed on the surface reacted with ZN-EL (Figure 1b). Moreover, energy-dispersive X-ray spectroscopy mapping confirmed that Zn was evenly dispersed (Figure S1). ZN-EL forms a Li–Zn alloy and LiF through a conversion reaction on the Li metal surface,²² and the Li–Zn alloy can function as a solid solution buffer layer during the Li plating/stripping process to induce uniform Li deposition. However, herein, the Li metal surface exposed to no-additive EL, N-EL, and LN-EL

did not exhibit notable changes (Figure S1), apart from a few particles formed because of the chemical reaction between the electrolyte and Li metal. Thus, the simple contact between ZN-EL and the Li metal induces in situ Li–Zn alloying, which probably results in the formation of a lithiophilic and stable SEI.

Figure 1c shows the voltage profile of the initial Li deposition process in a Li||Cu half-cell. This profile was used to evaluate the nucleation overpotential, which is the difference between the initial decrease in voltage during discharge and the voltage in the subsequent plateau region.^{27,28} At a low current density of 0.1 mA cm^{-2} , ZN-EL exhibited a nucleation overpotential of 15 mV (vs Li/Li⁺), which was more than twice as low as 37 mV (vs Li/Li⁺) observed with the no-additive EL. When Li–Zn sites with high solubility in Li metal are formed on the Li surface, the interfacial and nucleation energy barrier can be reduced due to their structural similarity to pure Li, which enables better lattice compatibility.²¹ Furthermore, even at a high current density of 5 mA cm^{-2} , a considerably lower nucleation overpotential was observed for ZN-EL (0.30 V, vs Li/Li⁺) compared with that for no-additive EL (0.39 V, vs Li/Li⁺) (Figure 1d).

To evaluate the stability of Li metal for each electrolyte, Li||Li symmetric cells were tested under a harsh current density condition of 5 mA cm^{-2} for no-additive EL, LN-EL, and ZN-EL (Figure 1e). Although the no-additive EL cell underwent a short circuit after ~250 cycles, the ZN-EL cell continued to operate stably for more than twice as many cycles. LN-EL slightly improved the lifespan of Li metal; however, a short circuit still occurred after an additional 50 cycles compared with no-additive EL. The superior long-term Li metal stability provided by ZN-EL is probably due to the formation of a Li–Zn alloy, which facilitates uniform Li plating and stripping, thereby mitigating dendrite growth. The voltage profiles shown in Figure S2 reveal that the three investigated electrolytes displayed peaks (marked with red circles), which corresponded to their increased overpotential in the discharge process. These peaks were attributable to pit formation or dendritic growth.⁶ Conversely, no such peaks were observed for ZN-EL,

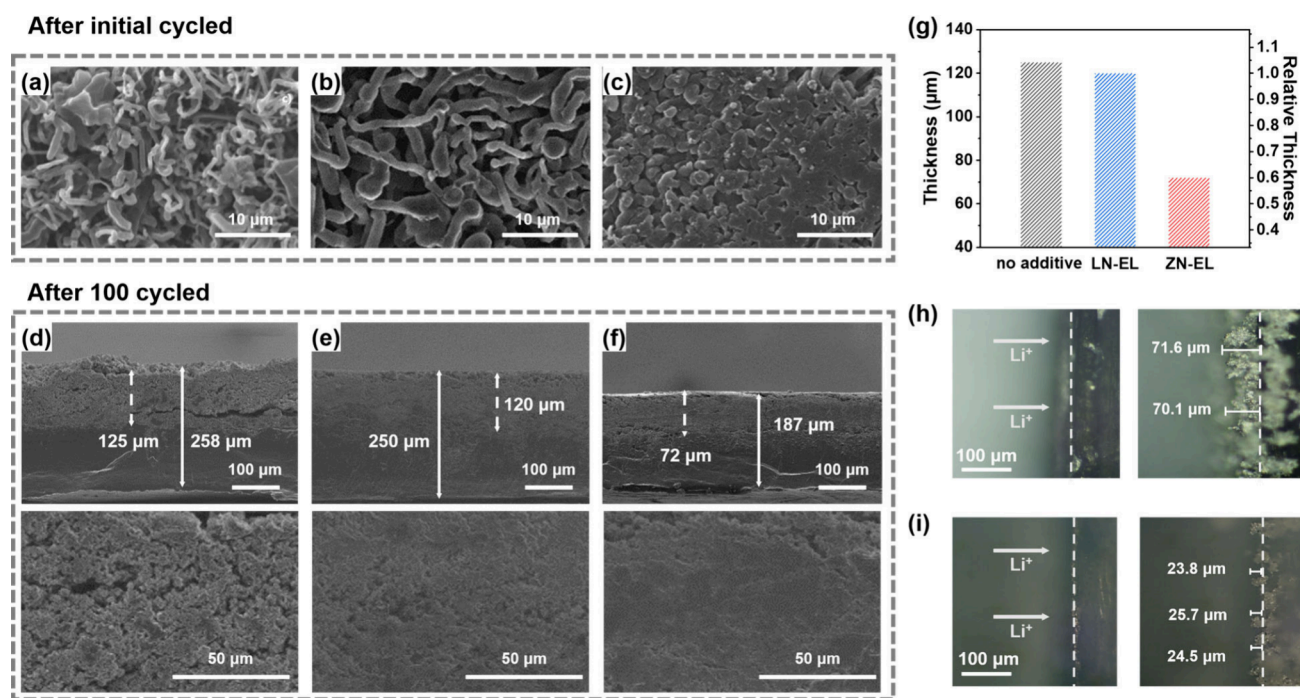


Figure 2. Comparison of the Li dendrite growth and deposition behavior. SEM images of the Li metal surface (a–c) after initial plating and cross sections (d–f) after 100 cycles, along with the corresponding enlarged cross-section images. The following electrolytes were utilized for plating: (a, d) 2.3 M LiFSI in DME (no-additive EL), (b, e) an electrolyte containing 80 mM LiF and 20 mM LiNO₃ (LN-EL), and (c, f) an electrolyte containing 40 mM ZnF₂ and 20 mM LiNO₃ (ZN-EL). (g) Bar chart of the real and relative thicknesses of the plated Li compared to the real and relative thicknesses of pristine Li metal. OM images of the initial deposition morphology of Li in the (h) no-additive EL and (i) ZN-EL. Li was deposited at a current density of 5 mA cm^{−2} up to a capacity of 0.5 mAh cm^{−2}.

confirming that Li deposition and stripping proceeded uniformly. Regarding the long-term cycling performance (Figure 1f), the initial overpotential was 32 mV (vs Li/Li⁺) for no-additive EL and ZN-EL and 38 mV for LN-EL. After 150 cycles, the overpotential increased to 48 mV for no-additive EL and LN-EL, whereas it increased to 46 mV for ZN-EL, maintaining the lowest overpotential at the 150th cycle. After 250 cycles, only the cell with ZN-EL retained an overpotential of 49 mV, whereas short circuits occurred in the other cells. In particular, the cell with ZN-EL exhibited more than twice the cycling lifespan of the cell with no-additive EL, achieving stable operation for over 500 cycles. Thus, the initial stripping/plating behavior affects the long-term lifespan of Li metal. When comparing the surfaces formed during the first cycle, a smooth surface was observed to exhibit a lower local current density than an uneven surface. This continuously suppressed dendrite growth, thereby increasing the lifespan of Li metal.²⁹

The initial deposition morphology and cross-section of Li metal exposed to the electrolytes were observed via SEM after 100 cycles. Figures 2a–c show the SEM images of the initially deposited Li metal surface plated with the no-additive EL, LN-EL, and ZN-EL. In the case of no-additive EL, Li exhibited a nonuniform and randomly grown shape, while LN-EL exhibited thicker dendrites compared with those observed for no-additive EL. Contrarily, ZN-EL exhibited the most uniform, bulk-like Li deposition. Figures 2d–f show cross-section images of the cycled Li (100 cycles), with thickness measurements numerically summarized in Figure 2g. Pristine Li metal had a thickness of 113 μm (Figure S3), while no-additive EL formed a thick porous layer (125 μm) due to the accumulated SEI residue and dead Li. LN-EL showed a slightly

thinner layer (120 μm) compared with no-additive EL, while ZN-EL exhibited a considerably thinner accumulated layer of 72 μm, a 60% reduction. Magnified SEM image of the porous layer for each electrolyte revealed that ZN-EL had the highest density, followed by LN-EL and no-additive EL, indicating that ZN-EL suppressed dendrite growth, resulting in a denser and less porous layer. The porous growth of Li metal and side reactions with the electrolyte led to continuous volume expansion. Based on this result, the per-cycle thickness increase was calculated to be 8.29% for additive-free EL and 7.97% for LN-EL. In contrast, ZN-EL exhibited the lowest increase at 5.05%, suggesting its effectiveness in mitigating volume expansion by suppressing dendritic growth (Table S1). Figure S4 shows that N-EL exhibited considerably thinner dendrites compared with LN-EL, which can be attributed to the presence or absence of LiF additives. In LN-EL, the increased amount of LiF enhanced the SEI's mechanical strength, effectively suppressing dendrite growth during deposition.³⁰

In situ optical microscopy (OM) was employed to investigate the initial Li deposition behavior and morphology. Figure 2h,i show the morphologies of the electrochemically deposited Li metal in no-additive EL and ZN-EL at 5 mA cm^{−2} until 0.5 mAh cm^{−2}. For no-additive EL, Li deposition was thin and irregularly shaped with varying thicknesses of 71.6 and 70.1 μm. Contrarily, ZN-EL enabled uniform Li deposition with thicknesses of 23.8, 25.7, and 24.5 μm, forming a thin and compact layer with a uniform thickness and structure. The lithiophilic Li–Zn alloy in ZN-EL facilitated smooth Li plating, reducing local current density and suppressing dendrite growth. Accordingly, the X-ray photoelectron spectroscopy and Li 1s depth profile analysis (Figure S5) confirmed the

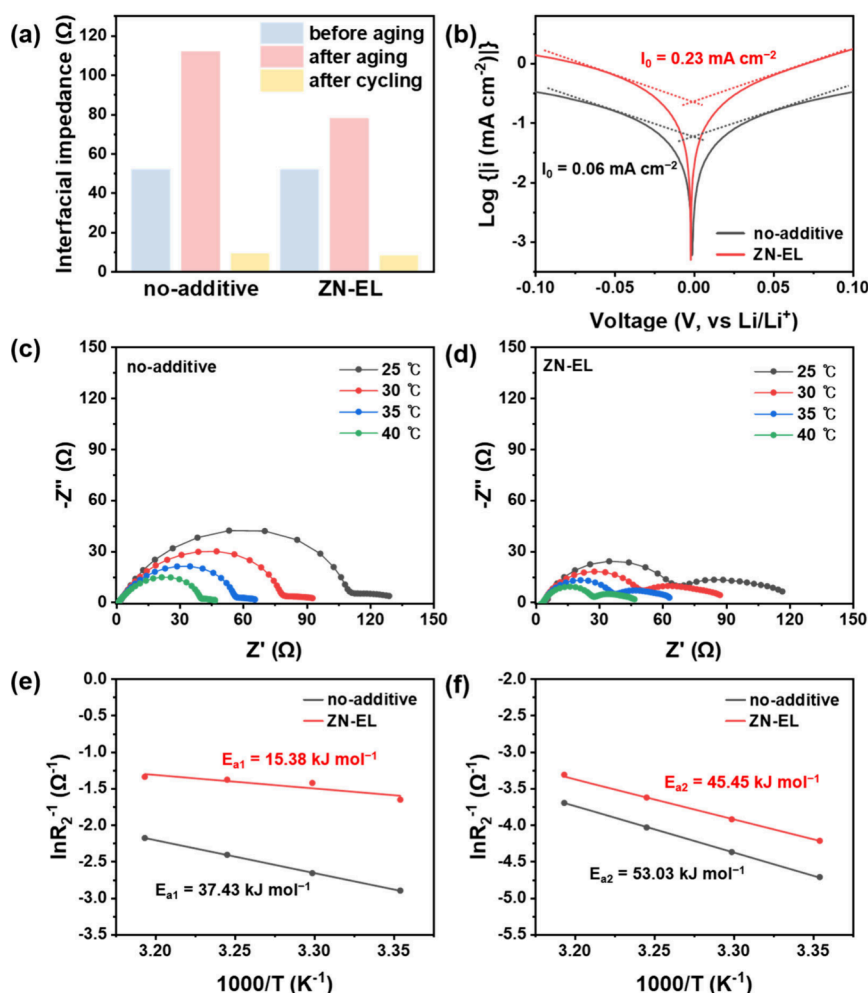


Figure 3. Electrochemical characterizations of Li||Li symmetric cells for interfacial kinetic behaviors. (a) Interfacial impedance obtained from the EIS spectra before and after aging, as well as after cycling. (b) LSV-obtained Tafel plots at 0.1 mV s^{-1} , along with the exchange-current densities calculated from the Tafel plot. (c, d) Nyquist plots of the Li||Li symmetric cells at different temperatures. The Arrhenius behavior, as well as the comparison of activation energies (e) E_{a1} and (f) E_{a2} between R_2 and R_3 derived from the Nyquist plots.

presence of a Li–Zn alloy on the surface of Li metal (after plating) and within the SEI. This alloy, along with Li_3N , enhanced the stability of the SEI and effectively suppressed dendrite growth. Li_3N , obtained via the electrochemical reduction of LiNO_3 , exhibits high ionic conductivity ($\sim 10^{-4} \text{ S cm}^{-1}$), which facilitates efficient Li transport through the SEI. This property offsets the inherently resistive nature of LiF (10^{-9} to $10^{-10} \text{ S cm}^{-1}$), yielding a more ion-conductive and electrochemically stable SEI structure.^{26,31} This phenomenon was further confirmed by electrochemical impedance spectroscopy (EIS) measurements, which revealed a notable decrease in the interfacial resistance in ZN-EL. Additionally, the results of the cathodic linear sweep voltammetry (LSV) analysis (Figure S6) revealed that ZN-EL suppressed the decomposition of FSI^- and DME, thereby minimizing side reactions with the electrolyte. These results indicated that ZN-EL offered mechanical and electrochemical stabilities, thereby reliably protecting the Li metal.

To reveal the advantages of the additives for electrochemical properties in Li metal, the interfacial stability was evaluated. Figure 3a shows the interfacial impedance values before and after aging (12 h) as well as after cycling in the Li||Li symmetric cells with no-additive EL and ZN-EL. Initially, both electrolytes had similar interfacial impedances, i.e., 52.01 and

52.54 Ω , but after aging, the impedance of both increased due to chemical reactions between Li metal and the electrolyte. ZN-EL, however, formed a Li–Zn alloy (through the conversion reactions described in eqs 1 and 2), exhibiting an interfacial impedance lower (78.0 Ω) than that for no-additive EL (112 Ω). After cycling, cells with no-additive EL and ZN-EL stabilized at 9.4 and 8.2 Ω , respectively. To evaluate the charge-transfer kinetics at the Li metal surface, LSV was performed and the exchange-current density (I_0) was determined using Tafel plots obtained from the LSV curves (Figure 3b). The higher the exchange-current density, the faster the charge transfer at the Li/electrolyte interface.^{32,33} The cell with no-additive EL had an exchange-current density of 0.06 mA cm^{-2} , whereas that with ZN-EL had an exchange-current density of 0.23 mA cm^{-2} . This indicates that the kinetics at the interface were improved in the cell with ZN-EL. Thus, the cell with ZN-EL, after aging, had a lower interfacial impedance compared with the cell with the no-additive EL (Figure 3a) owing to enhanced kinetics at the interface.

Considering that the charge-transfer process of Li ions across the SEI is dependent on temperature, the activation energy can be calculated by plotting the EIS spectra at various temperatures using the Arrhenius equation (Figures 3c–f).^{34,35} In the equivalent circuit model shown in Figure S7, R_{de} (R_2)

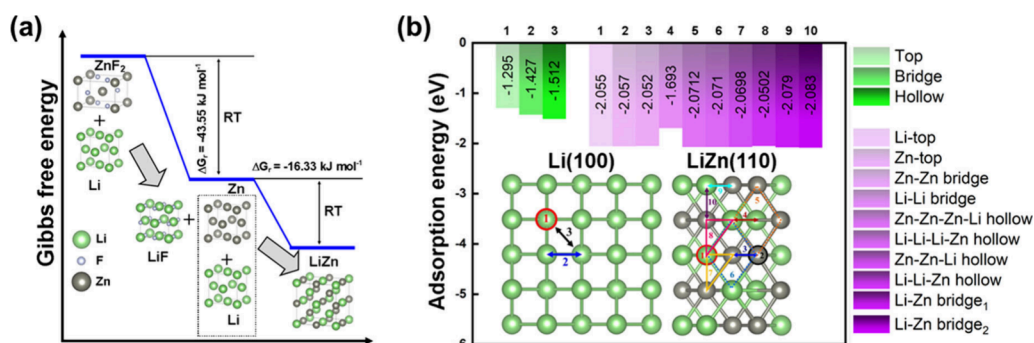


Figure 4. Computational calculations for the Li–Zn alloy. (a) Schematic illustration of the formation of LiF and LiZn along with their corresponding Gibbs free energy. (b) Li adsorption energies on Li (100) and LiZn (110) surfaces.

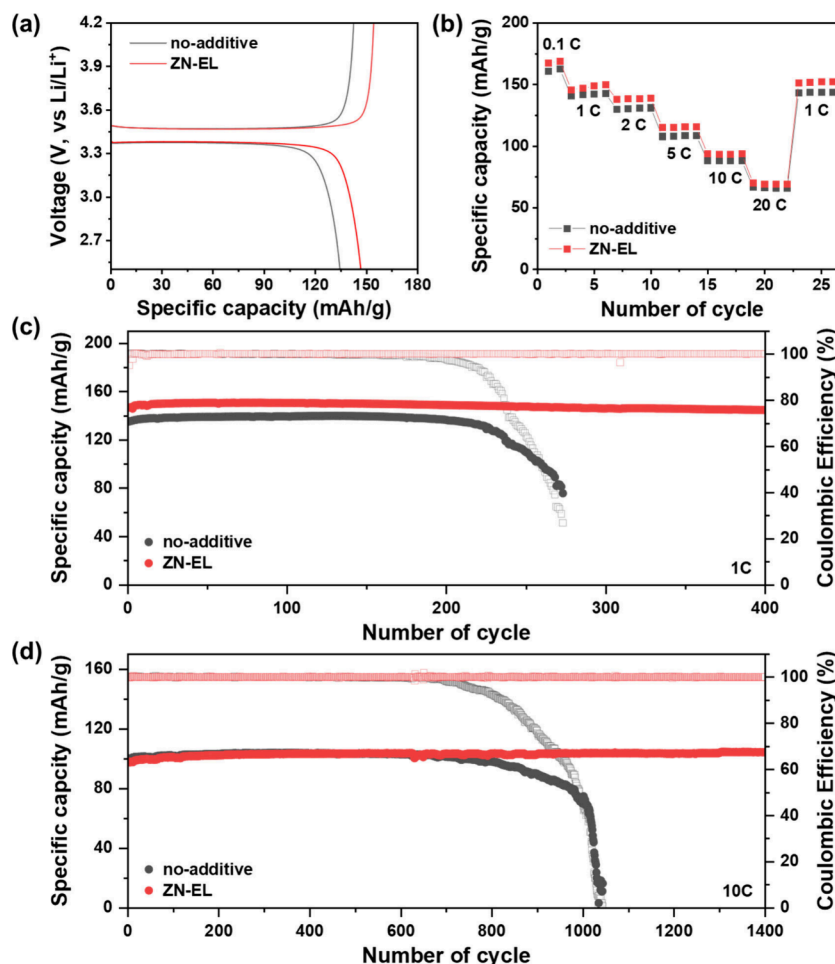


Figure 5. Electrochemical performances tests of Li||LFP cells. (a) Voltage profile for the 1st cycle at 1 C and (b) rate capabilities from 0.1 to 10 C. Long-term cycling stabilities at (c) 1 and (d) 10 C. All cells were tested in a voltage range of 2.5–4.2 V at 30 °C.

and R_{ct} (R_3) represent the resistances related to the desolvation of Li ions and charge transfer, respectively.³⁶ The activation energy (E_{a1}) for R_{de} in ZN-EL was significantly lower at 15.39 kJ mol^{−1}, compared with 37.43 kJ mol^{−1} for the cell with the no-additive EL. This suggests that the zinc cation positioned on the initial Li metal surface may have caused an increase in the local cation concentration. This facilitates the separation of solvent molecules during the desolvation of Li ions more easily than in the no-additive EL. The activation energy (E_{a2}) for R_{ct} in Zn-EL was reduced to 45.45 kJ mol^{−1} compared with 53.03 kJ mol^{−1} in no-additive EL. This energy barrier difference for

Li-ion transfer can be attributed to a stable SEI formation, comprising a lithiophilic Li–Zn alloy and Li₃N. The formation of the Li–Zn alloy in ZN-EL improved the interfacial stability and enhanced the kinetics of Li plating/stripping.

To verify the formation and effect of the Li–Zn alloy, density functional theory calculations were conducted using the Vienna Ab initio Simulation Package. Figure 4a illustrates the conversion reaction forming Li–Zn and LiF, with the Gibbs free energy (ΔG) calculations at 25 °C. The formation of Zn and LiF from ZnF₂ (eq 1) had $\Delta G = -43.55 \text{ kJ mol}^{-1}$, and the subsequent Li–Zn alloy formation (eq 2) had $\Delta G =$

$-16.33 \text{ kJ mol}^{-1}$, both indicating spontaneous reactions ($\Delta G < 0$). Therefore, ZnF_2 in ZN-EL absorbs onto the Li metal surface, forming the Li–Zn alloy, as shown in Figure 1.

Figure 4b shows the adsorption energies of Li ions deposited onto specific sites on the Li (100) and LiZn (110) surfaces; the details for each possible adsorption site are presented in Figures S8 and S9. The lowest adsorption energy on LiZn (110) (-2.083 eV) was significantly lower than that on Li (100) (-1.512 eV). Notably, even the highest adsorption energy on LiZn (110) (-1.693 eV) was still lower than the lowest adsorption energy on Li (100), indicating that Li was preferentially deposited onto the LiZn sites. In ZN-EL, ZnF_2 undergoes a conversion reaction to form a Li–Zn alloy, which serves as a nucleation site for uniform Li deposition, suppressing dendrite growth (Figure 2). This improved Li deposition sites, and kinetics in ZN-EL enhance electrochemical performance compared with that of the no-additive EL.

To verify the effect of the electrolyte additive in the full cell, the capacity and cyclability were evaluated using the LFP cathode. The first charge–discharge cycle curve was evaluated at 1 C ($1 \text{ C} = 160 \text{ mA g}^{-1}$) (Figure 5a).³⁷ The initial charge and discharge capacities of the cell with ZN-EL were close to the theoretical capacity (170 mAh g^{-1}) compared with the cell with no-additive EL. The cell with no-additive EL exhibited charge and discharge capacities of 142.6 and 134.6 mAh g^{-1} , respectively, whereas the cell with ZN-EL exhibited charge and discharge capacities of 154.4 and 146.7 mAh g^{-1} , with initial CEs of 94.4% and 95.0%, respectively. Figure 5b shows the rate capability assessed based on the discharge capacity at 1, 2, 5, 10, and 20 C, following two formation cycles at 0.1 C. The cell with ZN-EL exhibited higher capacities at each rate, recording 148.0, 138.6, 115.5, 93.6, and 69.4 mAh g^{-1} , while the cell with no-additive EL exhibited lower capacities of 141.9, 130.7, 108.4, 88.2, and 66.3 mAh g^{-1} , respectively. In addition, in the long-cycle evaluation at 1 C, the cell with the no-additive EL underwent capacity fading and a decrease in CE after approximately 200 cycles. Contrarily, the cell with ZN-EL maintained a capacity of 145 mAh g^{-1} even after 400 cycles, with an exceptionally high CE of 100% (Figure 5c). At this point, the cell with ZN-EL exhibited a retention of 98.6%, whereas the cell with the no-additive EL achieved the same retention at only 217 cycles. Considering that ZN-EL ensured a high current stability of the Li metal (Figure 1e), a high current was applied to a LillLFP full cell, and the cyclability was evaluated at an extremely high rate of 10 C (4.9 mA cm^{-2}) (Figure 5d). The cell with the no-additive EL began to decay after 800 cycles and rapidly failed after approximately 1000 cycles. Contrarily, the cell with ZN-EL maintained a capacity of 104.3 mAh g^{-1} even after 1400 cycles and demonstrated nearly 100% CE and retention at high rates, exhibiting an excellent long-term cycling performance. In the LillAl LSV test shown in Figure S10, the electrolyte in the cell with no-additive EL began to decompose at approximately 4.3 V (vs Li/Li^+), whereas in the cell with ZN-EL, decomposition occurred at approximately 4.7 V (vs Li/Li^+). This suggests that using electrolyte additives improves the stability of the anode and suppresses electrolyte decomposition at high voltages, reducing side reactions between the cathode and electrolyte.

In conclusion, we successfully improved the stability of Li metal in LMBs by introducing electrolyte additives ZnF_2 and LiNO_3 . These additives promoted the formation of a Li–Zn alloy on the Li metal surface and LiF and LiNO_3 as SEI

components. In the cell with ZN-EL, the lithiophilic Li–Zn alloy and reinforced SEI improved interfacial stability, suppressing dendrite growth, even at a high current density of 5 mA cm^{-2} . The effect of the additives was observed in the LillLFP full cell; at 10 C (approximately 5 mA cm^{-2}), the cell with ZN-EL exhibited a discharge capacity of 104.3 mAh g^{-1} after 1400 cycles with nearly 100% retention. This study demonstrates an electrolyte additive strategy that can considerably enhance the stability and lifespan of Li metal for next-generation high-energy-density LMBs through simple and effective treatment.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c01307>.

Additional data for FE-SEM images of the Li metal surface after reaction with electrolytes and after Li plating; voltage profiles of the symmetric cells, cross-sectional SEM images of pristine Li metals; cathodic and anodic LSV curves of electrolytes; calculation details for cumulative thickness increase per cycle (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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